

SolarGrid Demo Utility — Operating Reserve and Reliability Procedure

SG-RES-006

SOLARGRID AI — Synthetic Engineering Demonstration Corpus

Document ID	SG-RES-006
Document Title	SolarGrid Demo Utility — Operating Reserve and Reliability Procedure
Version	1.0
Revision Date	2026-01-15
Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY
Purpose	Define synthetic reserve calculations, reserve shortfall handling, corrective actions, and replanning triggers.

Demo Note: This document contains synthetic engineering requirements created for the SOLARGRID AI demonstration. All numerical values, thresholds, equipment ratings, and procedures are illustrative and must be replaced or validated by the deploying organization before operational use.

Use restriction: This document is intended only for RAG testing, semantic retrieval testing, SafetyAgent demonstration, engineering evidence demonstration, PlannerAgent integration testing, and Tool 16 testing. It is not intended for real-world grid operation.

Document Control

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Revision History

Revision	Date	Description	Author
1.0	2026-01-15	Initial synthetic demonstration release	SolarGrid Demo Utility — Engineering Department

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1. Purpose and Scope

This procedure defines the synthetic operating-reserve calculation and reliability workflow for the SolarGrid Demo Utility demonstration. It covers spinning reserve, non-spinning reserve, BESS reserve, generator reserve, reserve shortfall, corrective actions, and replanning triggers.

2. Reserve Definitions

RES-001 — Operating reserve is additional active-power capability that can be delivered within the planning response window while respecting equipment constraints.

RES-002 — Spinning reserve is upward capability from synchronized generators that can increase output without violating Pmax or ramp constraints.

RES-003 — Non-spinning reserve is capability from an available but not currently synchronized resource that can become deliverable within the defined response window.

RES-004 — BESS reserve is eligible upward discharge capability supported by BESS power, SOC, availability, and safety constraints.

3. Minimum Reserve Requirement

RES-005 — The minimum total operating reserve requirement is 25 MW.

RES-006 — Reserve calculations shall use the current operating snapshot rather than a historical or stale dispatch state.

4. Reserve Calculation Method

Total eligible reserve is the sum of qualifying generator reserve and qualifying BESS reserve, after applying all resource-specific constraints.

Resource	Raw Capability	Eligibility Checks
G1	Up to 25 MW	Headroom, ramp, synchronized state, availability
G2	Up to 20 MW	Headroom, ramp, synchronized state, availability
G3	Up to 15 MW	Headroom, ramp, synchronized state, availability
BESS	Up to 50 MW	SOC, reserve SOC, power limit, availability, safety

RES-007 — Reserve shall not be counted when the underlying resource is unavailable, restricted below the claimed capability, or blocked by a safety condition.

5. Satisfied Reserve Example

Example inputs: required operating reserve = 25 MW; available generator reserve = 18 MW; available battery reserve = 12 MW. Total available reserve = 30 MW.

RES-008 — The satisfied example has 30 MW total eligible reserve against a 25 MW requirement, leaving 5 MW of synthetic reserve margin.

6. Insufficient Reserve Example

Example inputs: required operating reserve = 25 MW; available generator reserve = 14 MW; available battery reserve = 6 MW. Total available reserve = 20 MW.

RES-009 — A total available reserve of 20 MW against a 25 MW requirement is a 5 MW reserve shortfall and shall trigger corrective action or replanning.

7. Corrective Actions

Priority	Action	Evidence to Recheck
1	Redispatch synchronized generators	Generator limits and ramp evidence
2	Adjust BESS dispatch/reserve	SOC, power, safety and network evidence
3	Commit another generator	Startup, minimum up/down and network evidence
4	Reduce forecast exposure / replan	Forecast and reserve evidence
5	Human review	Current system snapshot and approval status

8. Reserve and Network Interaction

RES-010 — Reserve satisfaction does not by itself prove network feasibility; reserve allocation shall be validated against voltage and thermal constraints.

Example: adding 25 MW of generator output capability may satisfy reserve numerically but could cause a transmission line to exceed 100% loading. In that case the reserve plan requires further engineering assessment.

9. Replanning Triggers

RES-011 — A reserve shortfall shall trigger replanning unless an approved operating exception explicitly changes the requirement.

RES-012 — A material forecast, generator availability, BESS availability, or network topology change shall trigger reserve reassessment.

10. Evidence Status

RES-013 — If reserve capability cannot be verified from current engineering evidence, reserve status shall be UNKNOWN rather than PASS.

This procedure is deliberately decomposed so RAG retrieval can connect a reserve calculation to generator evidence, BESS evidence, forecast evidence, and network evidence.